

An optimal acyclic-pivot certificate for the degree-seven Peterson hit matrix in three variables

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Abstract

For the degree-seven Peterson hit matrix over $\mathbb{F}_2$ in three variables, we prove that the maximum size of an acyclic pivot system on original labelled columns is $\tau(3, 7) = 26$. A complete support table exhibits 23 two-support columns forming four trees and three further columns with private sink pivots. Six zero rows and four parity constraints give the matching upper bound. This resolves the specified finite optimization and recovers the already-known matrix rank 26. The certificate allows arbitrary total row orders; no statement about other degrees or numbers of variables is asserted. Priority and broader novelty remain unverified.

1 The finite problem

Let $P = \mathbb{F}_2[x_1, x_2, x_3]$, with each generator of degree one. Write $x^a = x_1^{a_1} x_2^{a_2} x_3^{a_3}$ and $|a| = a_1 + a_2 + a_3$. The matrix $M = M(3, 7)$ has the 36 degree-seven monomials as rows and all original labels (q, a) with $q \in \{1, 2, 4\}$ and $|a| = 7 - q$ as columns. The column at (q, a) is the coefficient vector of

$$\text{Sq}^q(x^a) = \sum_{\substack{u_1, u_2, u_3 \geq 0 \\ u_1 + u_2 + u_3 = q}} \left(\prod_{j=1}^3 \binom{a_j}{u_j} \bmod 2 \right) x^{a+u}. \quad (1)$$

Here a binomial coefficient is zero when its lower argument exceeds its upper argument. Formula (1) is the usual unstable Steenrod action and Cartan formula. The label $q = 2^t$ simply replaces the label t . Zero columns are retained in M ; distinct labels would remain distinct even if their vectors coincided.

Following Phuc [1, Definition 4.12], an *acyclic pivot system* is a subset C of nonzero labelled columns and an injective map $p : C \rightarrow \{\text{rows of } M\}$ such that $p(c) \in \text{Supp}(c)$ and all arrows

$$r \longrightarrow p(c), \quad c \in C, \quad r \in \text{Supp}(c) \setminus \{p(c)\}, \quad (2)$$

form an acyclic directed graph. Let $\tau(3, 7)$ be the largest possible $|C|$. The columns must be original columns, without column operations. Total row orders need not be compatible with multiplication.

Theorem 1. *For this full 36×59 labelled matrix,*

$$\tau(3, 7) = \text{rank}_{\mathbb{F}_2} M(3, 7) = 26.$$

Equivalently, 26 is the maximum number of distinct support maxima of the nonzero columns over all total row orders.

The rank 26 and the lexicographic leader count 20 are already reported in [1, Example 6.3]. The equivalence with support-maxima optimization and the bound $\tau \leq \text{rank } M$ are [1, Theorem 4.14]. That work also discusses forest-based independence and private-pivot constructions. The contribution here is the explicit family attaining 26 for this full matrix across the Sq^1 and Sq^2 layers, with a complete support decomposition and a matching upper certificate. The inspected example does not state the value of $\tau(3, 7)$.

2 The complete support table

Index the row $x_1^a x_2^b x_3^{7-a-b}$ by

$$i(a, b) = 8a - \frac{a(a-1)}{2} + b, \quad 0 \leq a \leq 7, \quad 0 \leq b \leq 7-a. \quad (3)$$

These indices run from 0 to 35 in increasing exponent-lexicographic order. Table 1 lists every nonzero column, with a local index j ; the operation and source triple specify its original label uniquely. Its last column gives the pivots used below; a dash means omission.

There are $\binom{8}{2} = 28$, $\binom{7}{2} = 21$, and $\binom{5}{2} = 10$ source labels for $q = 1, 2, 4$, respectively. For $q = 1$, a degree-six source has either zero or two odd exponents. The ten all-even sources vanish and the other eighteen yield two terms each. For $q = 2$, the degree-five source exponent multisets are

$$(5, 0, 0), \quad (4, 1, 0), \quad (3, 2, 0), \quad (3, 1, 1), \quad (2, 2, 1).$$

The first two give nine zero columns. The last three give, respectively, six two-support, three four-support, and three two-support columns. Every $q = 4$ column vanishes because its source degree is $3 < 4$; higher powers have negative source degree. Thus the table contains all 30 nonzero labelled columns, and the remaining 29 columns are zero. For example, $\text{Sq}^2(x_1 x_2 x_3^3)$ has exponent triples $(1, 1, 5), (1, 2, 4), (2, 1, 4), (2, 2, 3)$, giving rows 9, 10, 16, 17.

Table 1: All nonzero original columns and the selected pivots.

j	q	Source triple	$\text{Supp}(c_j)$	Pivot
0	1	(0, 1, 5)	$\{1, 2\}$	2
1	1	(0, 3, 3)	$\{3, 4\}$	3
2	1	(0, 5, 1)	$\{5, 6\}$	6
3	1	(1, 0, 5)	$\{8, 15\}$	15
4	1	(1, 1, 4)	$\{10, 16\}$	16
5	1	(1, 2, 3)	$\{10, 17\}$	17
6	1	(1, 3, 2)	$\{12, 18\}$	12
7	1	(1, 4, 1)	$\{12, 19\}$	—
8	1	(1, 5, 0)	$\{14, 20\}$	20
9	1	(2, 1, 3)	$\{16, 17\}$	—
10	1	(2, 3, 1)	$\{18, 19\}$	18
11	1	(3, 0, 3)	$\{21, 26\}$	21
12	1	(3, 1, 2)	$\{23, 27\}$	23
13	1	(3, 2, 1)	$\{23, 28\}$	—
14	1	(3, 3, 0)	$\{25, 29\}$	25
15	1	(4, 1, 1)	$\{27, 28\}$	28
16	1	(5, 0, 1)	$\{30, 33\}$	33
17	1	(5, 1, 0)	$\{32, 34\}$	34
18	2	(0, 2, 3)	$\{2, 4\}$	4
19	2	(0, 3, 2)	$\{3, 5\}$	5
20	2	(1, 1, 3)	$\{9, 10, 16, 17\}$	9
21	2	(1, 2, 2)	$\{10, 12\}$	—
22	2	(1, 3, 1)	$\{12, 13, 18, 19\}$	13
23	2	(2, 0, 3)	$\{15, 26\}$	26
24	2	(2, 1, 2)	$\{16, 27\}$	27
25	2	(2, 2, 1)	$\{19, 28\}$	19
26	2	(2, 3, 0)	$\{20, 29\}$	29
27	2	(3, 0, 2)	$\{21, 30\}$	30
28	2	(3, 1, 1)	$\{23, 27, 28, 31\}$	31
29	2	(3, 2, 0)	$\{25, 32\}$	32

3 The matching certificates

First, every acyclic pivot system with m columns has a nonsingular $m \times m$ minor. Indeed, choose a topological order of the graph in (2), so each nonpivot support row precedes its pivot. Sort the selected columns and their distinct pivot rows by increasing pivot position. The resulting square submatrix has diagonal entries one. An entry below the diagonal is zero, since a later pivot row cannot occur in a column with an earlier pivot. The minor is unit upper triangular, proving $m \leq \text{rank } M$.

For the rank upper bound, partition the rows into

$$\begin{aligned} Z &= \{0, 7, 11, 22, 24, 35\}, \\ B_1 &= \{1, 2, 3, 4, 5, 6\}, \\ B_2 &= \{8, 15, 21, 26, 30, 33\}, \\ B_3 &= \{9, 10, 12, 13, 16, 17, 18, 19, 23, 27, 28, 31\}, \\ B_4 &= \{14, 20, 25, 29, 32, 34\}. \end{aligned}$$

Every nonzero column in Table 1 is supported in one B_i and has even support size. Consequently every vector v in the column span satisfies

$$v_z = 0 \ (z \in Z), \quad \sum_{r \in B_i} v_r = 0 \ (1 \leq i \leq 4).$$

These ten linear functionals have pairwise disjoint, nonempty supports and are therefore independent: evaluating at one coordinate in any support isolates its coefficient. Their common kernel has dimension $36 - 10 = 26$. Hence

$$\tau(3, 7) \leq \text{rank } M \leq 26. \quad (4)$$

For the lower bound, regard each of the 27 two-support columns as an undirected edge. Delete columns 7, 9, 13, 21. The remaining 23 edges are exactly the four disjoint trees

$$\begin{aligned} T_1 &: 1 - 2 - 4 - 3 - 5 - 6, \\ T_2 &: 8 - 15 - 26 - 21 - 30 - 33, \\ T_3 &: 17 - 10 - 16 - 27 - 28 - 19 - 18 - 12, \quad \text{with the additional edge } 27 - 23, \\ T_4 &: 14 - 20 - 29 - 25 - 32 - 34. \end{aligned}$$

Their vertex counts are 6, 6, 9, 6, and their edge counts sum to $5 + 5 + 8 + 5 = 23$. Root them at 1, 8, 10, 14, respectively. Pivot each edge at its endpoint farther from the root. Each nonroot vertex is the pivot of its unique parent edge, so these 23 pivots are distinct; all arrows point away from a root and are acyclic.

The remaining three columns 20, 22, 28 have supports

$$\{9, 10, 16, 17\}, \quad \{12, 13, 18, 19\}, \quad \{23, 27, 28, 31\}.$$

Choose their pivots to be 9, 13, 31, respectively. Each of these rows appears in exactly its own column in the full nonzero table. The other support rows belong to T_3 . The added pivots are thus distinct new sinks, and adding all arrows into them cannot create a cycle. This gives an acyclic system of $23 + 3 = 26$ original labelled columns.

For an explicit check, the table's pivots are compatible with the following total row order, written from lowest to highest and read across successive lines:

$$\begin{aligned} &0, 7, 11, 22, 24, 35, 1, 2, 4, 3, 5, 6, \\ &8, 15, 26, 21, 30, 33, 10, 16, 17, 27, 23, 28, \\ &19, 18, 12, 14, 20, 29, 25, 32, 34, 9, 13, 31. \end{aligned} \quad (5)$$

Therefore $26 \leq \tau(3, 7)$, which with (4) proves both equalities in Theorem 1. Finally, for any total order, selecting one column per distinct support maximum gives an acyclic pivot system. Conversely, a topological order of any such system realizes all its pivots as distinct support maxima. This proves the stated equivalence as well. $\square$

4 Scope and reproducibility

The certificate settles only the specified finite optimization. It does not assert $\tau = \text{rank } M$ for other matrices or resolve the general Peterson hit problem. The rank value and the elementary forest principle are prior results; the attained acyclic-pivot optimum is the finite quantity determined here. The literature comparison uses the retained version 2 of [1] and a bounded source screen dated 8 September 2026. Priority, global duplication, and broader novelty remain unverified.

As broad adjacent computational context, Zeng [2] studies a finite monomial-algebra problem with exact matrix certificates. That reference is not used to establish any assertion about Steenrod columns or the value of $\tau(3, 7)$.

All coefficients and certificates above are finite and exact. The proof requires only formula (1), the support table, topological sorting, and elementary linear algebra. The accompanying standard-library Python script `verify.py`, run from the source directory as `python3 verify.py`, reconstructs all 59 labelled columns by the Cartan expansion and checks the printed support and pivot table, all constraints in (5), the triangular minor, and the ten annihilators. No floating-point arithmetic or optimization solver is needed for verification.

Computational and AI assistance. OpenAI Codex assisted with the initial permutation/swap search, the structural certificate, exact verification code, and manuscript drafting. A separate model-context mathematical audit independently regenerated the matrix by choosing degree-one factors to square and checked the certificates. This was an additional computational review, not external peer review or a formal proof-assistant verification. The complete finite argument is given above; search outcomes and model judgments are not premises of the proof.

References

- [1] Dang Vo Phuc, *Combinatorial Bounds on the Peterson Hit Problem via Certified Matrix Minors*, arXiv:2608.02623v2 (2026). <https://arxiv.org/abs/2608.02623v2>.
- [2] Zijian Zeng, *The Whiskered-Graph Weak Lefschetz Conjecture for Graphs on Eight Vertices* (2026). <https://doi.org/10.2139/ssrn.7385138>.